Course Content - PySpark

Day 1

1. Database:

What is the ER Model?
Why is ER Modelling neccesary?
MySQL Overview
Queries
DDL
Queries
DML Queries
Select Queries
Group By, Order By
Clause Date Functions in
SQL Integrity Constraints
Keys [Primary Key, Foreign Key, Composite Key]
Using Joins
Working with Windows Function
Understanding difference in Windows Function vs Group By Function
Optimization of Queries
Using Indexes and Triggers
Understanding the concept of ETL as well as Data Warehousing
Use of intermediate or staging table during ETL

2. HDFS:

Understanding the problem statement and challenges persisting to large data and how HDFS solves that problem,
Understanding the HDFS architecture
Blocks
Replication Factor
Rack awareness
New implementation in Hadoop 3.0 in context to HDFS
HDFS Erasure Coding
Exploring the HDFS using Command-line as well as Web Interface[Ambari]
Writing files to HDFS
File Formats created for Hadoop [Avro,Parquet,ORC]
File Formats created for Hadoop [Avro,Parquet,ORC]
Avoiding Single Point of Failure in HDFS
Understanding Read and Write Pipelines in HDFS
HDFS commands
Avoiding Single Point of Failure in HDFS
Using HDFS Federation
Using ZooKeeper

Day 2

3. MapReduce and YARN:

Understanding how the distributed processing solves the big data challenge
and how MapReduce helps to solve that problem,
Word count problem and solution,
MapReduce flow,
Explain the Driver,
Mapper and Reducer
code, Testing,
debugging project through eclipse and then finally packaging,
deploying the code on Hadoop Cluster,
Input Formats - Input splits & records,
text input,
binary input,
Output Formats - text output, binary output, lazy output,
MapReduce combiner,
MapReduce partitioner,
Data locality,
Speculative execution,
Job optimization.
DataSet Solution for :
https://openflights.org/data.html Classical version of
Apache Hadoop (MRv1) Limitations of classical
MapReduce
Addressing the scalability
resource utilization issue and need to support different programming paradigms,
YARN: The next generation of Hadoop's compute platform (MRv2)
Architecture of YARN
Application submission in YARN
Type of yarn schedulers (FIFO, Capacity and Fair).
New Features in Hadoop 3.0
Case Study on Aadhar Card Data

Day 3

4. Hive

Hive Architecture and components with typical query flows
Creating the table
Loading the datasets & performing analysis on that Datasets
Types of Hive metastore configurations
Understanding Hive Data model
Running the DML commands like Joining tables
writing sub query
saving results to table or HDFS etc.

Understanding the different File formats and choosing the right one
when to use partitioning
bucketing to optimize query performance
Writing UDF's to reuse project/Domain specific implementations
Loadind data in Hive Table from Regex,XML and JSON Format
Recap: Candidates need to study Regex and structure of JSON and XML
Indexes in Hive
Date Functions in Hive
Window Functions in Hive
Coverting Data in ORC and Parquet Format in Hive
Data set that will be used
:https://data.cityofchicago.org/Transportation/Taxi-Trips/wrvz-psew

Day 4

5. HBASE:

Understanding the Need of NoSQL Database
NoSQL vs RDBMS
CAP Theorom
Amazon Redis vs MongoDB vs HBase vs GraphQL
Understanding the complete architecture of HBASE
Region Server
HBase Master
ZooKeeper
Moodeling the data into column families
Writing the queries to interact with the data stored in HBASE.
Performing HBASE-HIVE Integration

Day 5

6. SQOOP:

Setting up RDBMS Server and creating & loading datasets into RDBMS Mysql
Sqoop Architecture
Writing the Sqoop Import Commands to transfer data from RDBMS to HDFS/Hive/Hbase
Incremental Imports
Writing the Sqoop Export commands to transfer data from HDFS/Hive to RDBMS
Sqoop Jobs to automate those sqoop commands for day to day use

7. Spark Core:

Understanding Disk Computing vs In Memory Computing
How Spark works
Understanding the Spark architecture and why it is better than Map Reduce
RDD: Unit of Data in Spark
Architecture of Spark
Understanding SparkContext, Worker Nodes, Executioner and Tasks

Day 6

How Spark supports multiple languages
How beneficial is Unified Spark API
Tranformation and Actions in RDD
How DAG's are formed
How does Spark Lazy Loading works
Broadcast, Accumolator in Spark Core
Persiting and Caching Data in Spark
DAG scheduler
physical plan, logical plan
Common problems
Shuffles, Spills, small files
Optimizations
Spark Tuning
Memory Tuning
Garbage Collection
Tuning Data Structure
Tuning

Day 7

8. Spark SQL:

Why Spark SQL?
Understanding Spark SQL behind the hoods working
How Spark RDD still forms the base of Spark SQL
Data Frame : Basic Unit of Data in Spark SQL
Dataset : TypeSafety unit of data in SparkSQL for Object Oriented Languages like Java and scalability
Loading Data from following format into Datasets or DataFrames
1. JSON
2. CSV
3. Parquet
4. ORC
Using Joins in Spark
Connecting Spark with Hive
Connecting Hive with HBase
Exporting Data from Spark

Migration for Spark 2.6 to 3.0
New Features in Spark
Adaptive Query Execution
Dynamic Partition Pruning
Java 11 support
Hadoop 3 support
Delta Table from fast Execution

Case Study using Spark SQL

Day 8

9. Oozie:

Oozie Fundamentals
Oozie workflow
creations Oozie Job
submission monitoring
debugging using oozie web console and command line,
Concepts on Coordinators and Bundles

10. Logging and Alerts:

Log4j Properites
HDFS alerts
HDFS high availability alerts
NameNode high availability alerts
YARN alerts
MapReduce2 alerts
HBase service
alerts Oozie alerts
ZooKeeper alerts

Day 9

11. Data Governance:

What is Data Governance?
What is Data Security?
Using Kerberos

12. Understanding Administrator Framework/Architecture:

Components
High
Availability
NameNode, Resource Manager
Hive
HBase HA reads, HA tables, cluster capacity,
sizing Configurations
configuration.properties
yarn-site.xml,
mapred-site.xml
hbase-site.xml
hive-site.xml

Day 10

13. Creating an ETL Pipeline:

Pipeline concepts
Source types
DB
Flatfiles (S3)
APIs
HDFS
Destination
types DB
HDFS
Load types
batch / stream
full loads, incremental loads
Scheduling
Logging
Monitoring
Alerting
Quality checks
Backfills and Refills
Making an ETL Pipeline using Hive
Making an ETL Pipeline using Spark